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Hericium erinaceus extracts alter behavioral rhythm in mice

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ABSTRACT

Hericium erinaceus (HE), an edible mushroom, has been used as a herbal medicine in several Asian countries since ancient times. HE has potential as a medicine for the treatment and prevention of dementia, a disorder closely linked with circadian rhythm. This study investigated the effects of the intake of HE extracts on behavioral rhythm, photosensitivity of the circadian clock, and clock gene mRNA expression in the suprachiasmatic nucleus (SCN), a central clock, in mice. Although the HE ethanol extract only affected the offset time of activity, the HE water extract advanced the sleep–wake cycle without affecting the free-running period, photosensitivity, or the clock gene mRNA expression in SCN. In addition, both extracts decreased wakefulness around end of active phase. The findings of the present study suggest that HE may serve as a functional food in the prevention and treatment of Alzheimer's disease and delayed sleep phase syndrome.

Hericium erinaceus (HE), an edible mushroom, has been used as a herbal medicine in several Asian countries since ancient times. HE has been reported to possess antitumor properties. Intraperitoneal administration of HE extracts to tumor-bearing mice for a 2-week period significantly reduced tumor size through the induction of NK cell activity, activation of macrophages, and inhibition of angiogenesis (11). Polysaccharides from HE also enhanced apoptosis induced by doxorubicin, a chemotherapeutic drug, in human hepatocellular carcinoma cells (13). Compounds isolated from the fruiting body of HE also reduced proliferation of HL-60 leukemia cells through induction of apoptosis (14).

In addition, HE has been reported to have beneficial effects on the central nervous system, with HE intake found to reduce depression and anxiety in women (21). Hericenones isolated from the fruiting body and erinacines isolated from the mycelium of HE promoted nerve growth factor (NGF) biosynthesis in rodent astrocytes cultures (15). HE also prevented cognitive deficits induced by intracerebroventricular administration of amyloid β (25–35) peptide in mice (18). HE has been found to be effective in improving mild cognitive impairment in humans, which might be attributable to NGF synthesis promoting effect by hericenones (16). Repeated treatment with HE had neuroprotective effects in mice that were subjected to middle cerebral artery occlusion, through promotion of NGF synthesis (3). These studies suggest that HE has potential as a medicine for treatment and prevention of dementia.

Dementia is closely linked with circadian clock, an internal clock that generates approximately 24 hours rhythm called as the circadian rhythm, and patients with dementia show functional circadian clock impairments (26). Although some studies have demonstrated that HE might be effective in the treatment and prevention of dementia, no previous studies have

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focused on the effect of HE on circadian clock disturbance.

In the present study, we examined the effects of the intake of HE extracts on behavioral rhythm in mice. We also investigated the effects of HE extract intake on photosensitivity of the circadian clock and clock gene mRNA expression in the suprachiasmatic nucleus (SCN), a central clock that controls circadian rhythm.

MATERIALS AND METHODS

Preparation of HE extracts. Freeze-dried fruiting bodies of HE were obtained from Aso Biotech Co., Ltd. (Kumamoto, Japan). The freeze-dried fruiting bodies were agitated in ethanol for 24 h before being filtered. Ethanol extraction of the residue was repeated two more times. The filtrates were evaporated and dried in a vacuum desiccator. The final product was an ethanol extract (EE). The ethanol extraction residue was agitated in ultrapure water for 4 h and centrifuged at 3000 rpm for 10 min. The supernatant was filtered and the water extraction procedure was repeated. The filtered supernatant was lyophilized. The final product was the water extract (WE).

Animals and treatments. Five-week-old male CBA/N mice were purchased from Japan SLC, Inc. (Shizuoka, Japan). Mice were maintained under a 12 : 12 h light–dark cycle (LD condition) at 23 ± 2°C. Food and water were available *ad libitum*. Mice were housed under these conditions for 1 week before the start of the experiments. After this preliminary period, mice were fed MF powder (Oriental Yeast Co., Ltd, Tokyo, Japan) containing the HE extract (2% EE or 2% WE). At 7–15 days after the start of the diet containing the HE extract, behavioral rhythm was measured in a 12 : 12 h LD condition. Next, behavioral rhythm was also measured in a 12 : 12 h dark–dark cycle (DD condition) over a 2-week period. At 7 days after the start of housing in the DD condition, a light pulse (15 min) was delivered at circadian time (CT) 15 (i.e., 3 h after the onset of activity in DD, defined as CT 12). At 1 week after the light pulse, mice were returned to the LD condition for 1 week, after which the 15 min light pulse was delivered at zeitgeber time (ZT; ZT 0 represents lights-on) 15. An hour after the light pulse, the mice were euthanized and the clock gene mRNA expression levels in SCN were determined. In addition, to investigate time-course changes of clock gene mRNA expression, another series of mice were fed MF powder (Oriental Yeast Co., Ltd) containing HE extract (2% WE) after a preliminary period. At 2 weeks after the start of the WE-containing diet in LD condition, the mice were euthanized at ZT 0, 4, 8, 12, 16, and 20 and the clock gene mRNA expression levels in SCN were determined. Animal care and experimental procedures were performed in accordance with institutional guidelines, with the approval from the Kyushu University Institutional Review Board.

Measurement of behavioral rhythm. At 1 week after the start of the intake of diets containing the HE extracts, measurement of behavioral rhythm was initiated. Mice were individually housed in cages equipped with running wheels. Running-wheel activity was automatically recorded with a computer system (Chronobiology Kit; Stanford Software Systems, Palo Alto, CA). ClockLab software (Actimetrics, Evanston, IL) was used to determine the rhythm profiles, free-running period, and light-induced phase shift in the mice fed HE extract.

Analysis of clock gene mRNA expression levels. The mice were euthanized, and their brains were removed. The mRNA levels of clock genes (*Per1*, *Per2*, *Dbp* and *Bmal1*) in SCN were determined by *in situ* hybridization, as described by Iwamoto *et al.* (5). Coronal sections of the SCN (18 μm thick) were prepared using a cryostat. Antisense 45-mer oligonucleotide probes for *Per1* (antisense sequence: 5'-GTCCCTGG TGCTTTACCAGATGCACATCCTTACAGATCTG CTGGA-3'), *Per2* (5'-GCAGTCACTGCCCTGAGA GTCCCGTCCCGTGGAGCAGTTCTCGTT-3'), *Dbp* (5'-CAGGTCACCACGAGTCAGGAAACAG GCAACCGCGTGTCTGGCTAA-3'), and *Bmal1* (5'-GCCATTGCTGCCTCATCGTTACTGGGACTA CTTGATCCTTGGTCG-3') were labeled with [^{33}P]-dATP (PerkinElmer NEW® [New England Nuclear], Boston, MA). Hybridization was carried out overnight at 42°C. After glass slides were washed, they were air-dried and exposed to an X-ray film (Kodak, Tokyo, Japan) for 2 weeks with ^{14}C-labeled standards (American Radiolabeled Chemicals, St Louis, MO). The relative optical density was measured using a computerized image-analysis software (ImageJ; National Institute of Health, Bethesda, MD) and converted into the radioactive value (nCi) using the ^{14}C-labeled standard measurement. Two sections, including the intermediate portion of SCN, for each gene were selected from every animal, and whole areas of SCN (left and right, separately) were analyzed from each section. Left and right SCN signals were averaged, background signals were subtracted, and sig-

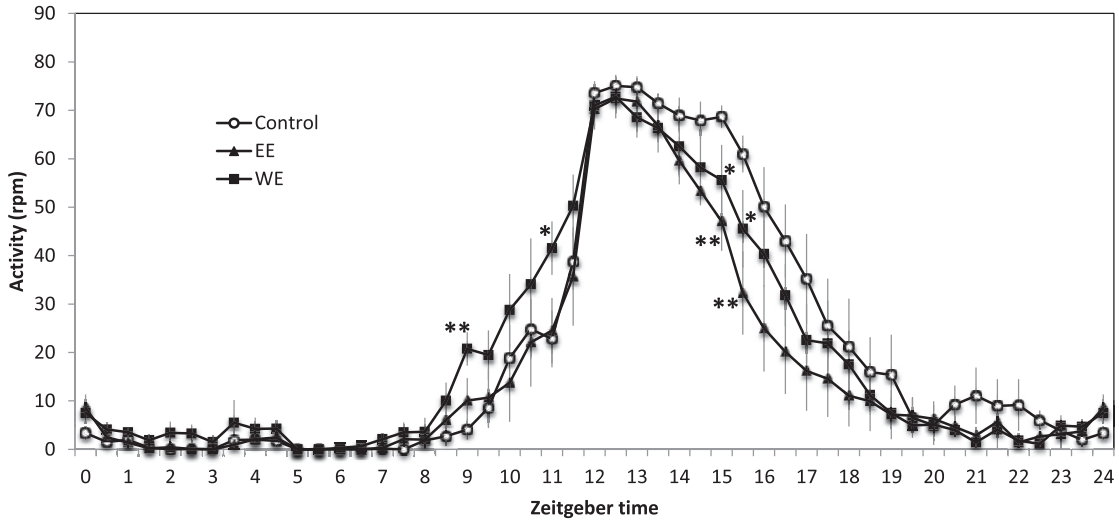

Fig. 1 The effect of ethanol extract (EE) and water extract (WE) of *Hericium erinaceus* (HE) on the behavioral rhythm in the light–dark (LD) condition. Values are expressed as mean ± standard error of the mean (SEM), $n = 6$ per group. $^{*}P < 0.05$ and $^{**}P < 0.01$ indicate significant differences from the control group.

nals for two sections were averaged for each animal.

Statistical analysis. Values were expressed as mean ± standard error of the mean (SEM). Data were analyzed using the two-way ANOVA, followed by a Tukey–Kramer test for multiple comparisons.

RESULTS

Effect of HE extract intake on behavioral rhythm

In the LD condition, the running-wheel activities of the control group began to increase at approximately 2 h before the start of the dark phase and began to decrease approximately 4 h after the start of the dark phase (Fig. 1). In contrast, the running-wheel activities of the EE and WE groups began to increase approximately 3 h before the start of the dark phase. The WE group showed significantly higher values than the control group at ZT 9 ($P < 0.01$) and ZT 11 ($P < 0.05$). The running-wheel activities of the WE and EE groups began to decrease approximately 2 h after the start of the dark phase and were significantly lower than those of the control group at ZT 15 (EE: $P < 0.01$, WE: $P < 0.05$) and ZT 15.5 (EE: $P < 0.01$, WE: $P < 0.05$).

The control group showed slight activity in the resting phase after the running-wheel activities (around ZT 20–23), whereas the WE and EE groups did not show any activity during the resting phase.

Effect of HE extract intake on photosensitivity

No significant differences among the groups were observed in the free-running period in the DD condition, and the light pulse-induced phase shift (Table 1). In the control group, light pulse increased mRNA levels of *Per1* and *Per2* in SCN (Fig. 2). The *Per1* and *Per2* mRNA levels in SCN of EE and WE groups also increased following the light pulse. There were no significant differences among the groups regarding *Per1* and *Per2* mRNA levels following the light pulse. Therefore, in the LD condition, HE extract intake did not affect the clock gene mRNA expression levels in SCN after the light pulse at ZT 15.

Effect of WE intake on clock gene mRNA expression levels

Because WE intake affected the sleep–wake cycle, which is also disturbed in patients with AD (22), we focused on the effects of WE intake on the clock gene mRNA expression levels in SCN. The mRNA levels of *Per1*, *Per2*, *Dbp*, and *Bmal1* in SCN were almost the same in the control and WE groups at any time point (ZT 0, 4, 8, 12, 16, and 20) in the LD condition (Fig. 3). No significant differences between the control and WE groups were observed. Thus, WE intake did not affect the clock gene mRNA expression levels in SCN at any time point in the LD condition.

DISCUSSION

The present study investigated the effects of the intake of HE extracts on behavioral rhythm, photosensitivity of the circadian clock, and clock gene mRNA

Table 1 *The effect of HE extract on the free-running period and the light pulse-induced phase shift in the dark–dark (DD) condition*

	Free-running period (h)	Phase-shifting (h)
Control	22.278 ± 0.169	−1.680 ± 0.292
EE	22.183 ± 0.107	−2.197 ± 0.281
WE	22.274 ± 0.113	−1.978 ± 0.349

Values are expressed as mean ± SEM, $n = 6$ per group.

expression levels in SCN.

We found that WE intake moved the onset and offset time of the activity forward, and EE moved the offset time of the activity forward. Both WE and EE decreased wakefulness around end of active phase. Conversely, the free-running period in the DD condition was not altered by WE or EE intake. In addition, WE and EE did not affect the phase shifting response or clock gene mRNA expression levels in SCN after the light pulse. Moreover, WE intake did not affect the clock gene mRNA expression levels in SCN at any time point in the LD condition. Therefore, changes of the behavioral rhythm following WE and EE intake might not be attributable to the effects of WE and EE on the properties of the circadian clock itself, i.e. free-running period, photosensitivity, or clock gene mRNA expression levels in SCN. Alternately, WE and EE might affect downstream pathway of the circadian clock, which links the circadian clock and behavioral output.

Several previous studies have reported that hericenones (7, 8) and erinacines (9, 10, 12, 23), chemical constituents of HE, stimulated the synthesis of NGF, which is the neurotrophic factor promoting the differentiation and survival of neurons. However, NGF injections into SCN caused phase shifts of the activity rhythm in hamsters kept in constant darkness (1). This implies that the alteration of behavioral rhythm following HE extract intake may be associated with NGF in SCN. However, further studies are needed in order to elucidate the relationship between the alteration of behavioral rhythm and NGF levels in SCN following HE extract intake because we did not measure the changes in NGF levels following HE extract intake.

In addition to the stimulation of NGF synthesis by hericenones (7, 8) and erinacines (9, 10, 12, 23), Nagai *et al.* (20) found that dilinoleoyl-phosphatidylethanolamine from HE protected neuronal cells from cell death caused by amyloid-β (Aβ)-induced endoplasmic reticulum stress. In the brains of mouse models of AD, erinacine A and erinacine S, structural analogs included in erinacines isolated from the mycelium of HE, attenuated the Aβ plaque burden through increasing Aβ degradation by elevating the level of insulin-degrading enzyme (2). HE has also been reported to be effective in improving mild cognitive impairment (16). In addition, hericenone B, one of hericenones isolated from the fruiting body of HE, was found to inhibit collagen-induced platelet aggregation, which is responsible for thrombosis leading to vascular dementia (17). Therefore, HE may be useful in producing a supplement for the treatment and prevention of dementia.

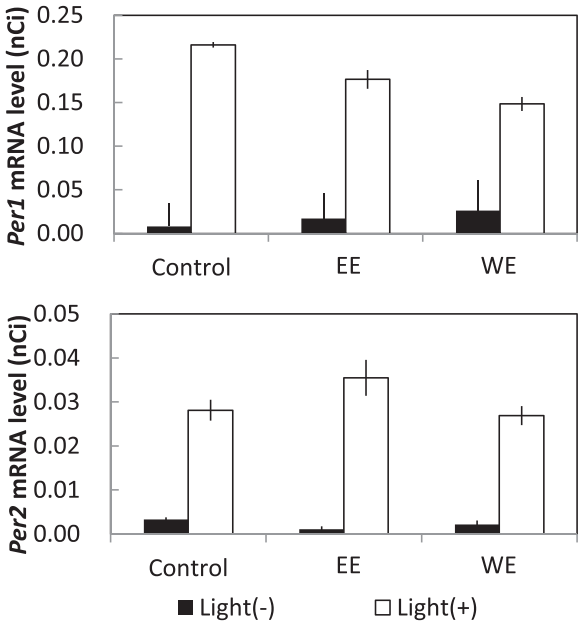

Fig. 2 The effect of HE extract on clock gene mRNA expression levels in the suprachiasmatic nucleus (SCN) following the light pulse at zeitgeber time 15 in the LD condition. Values are expressed as mean ± SEM, $n = 6$ per group.

Patients with AD show sleep–wake cycle disturbance, with frequent awakenings during the night and excessive daytime napping (22). Kang *et al.* (6) demonstrated that Aβ levels in the brain interstitial fluid were correlated with wakefulness and significantly increased during acute sleep deprivation. It has been suggested that sleep–wake cycle disturbance is not only a consequence of the disease process in AD but may also precede symptom onset and drive disease pathology (19). In the present study, WE and EE decreased wakefulness around end of active phase, suggesting that these HE extracts may shorten the latency to sleep onset and normalize the sleep–wake cycle in patients with AD. Therefore, WE and EE may be useful in the prevention and treatment of AD.

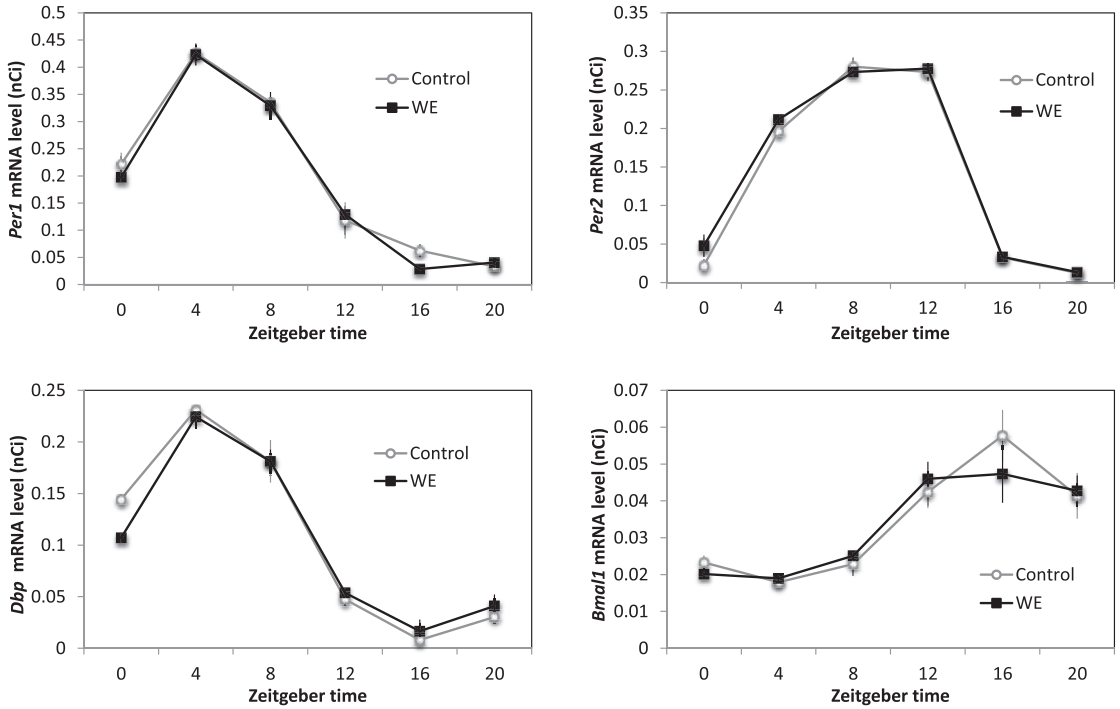

Fig. 3 The effect of WE intake on the clock gene mRNA expression levels in SCN in the LD condition. Values are expressed as mean ± SEM, $n = 4$ per group.

Delayed sleep phase syndrome (DSPS) is a circadian rhythm sleep disorder characterized by marked difficulties in initiating sleep at a conventional bedtime and in waking in the morning (25). It usually develops during adolescence, but childhood occurrence and persistence into adulthood have been reported. Our findings suggest that WE and EE intake caused the advance of the sleep phase in adolescent mice. Therefore, WE and EE intake may lead to improvement in DSPS.

HE extracts have also been reported to have an anti-obesity effect (4). Ethanol and hot-water extracts of HE were found to improve lipid metabolism in mice fed a high-fat diet, and the effect of the ethanol extract was mediated, at least in part, by activation of peroxisome proliferator-activated receptor α (PPARα) (4). Shirai *et al.* (24) demonstrated that bezafibrate, an anti-hyperlipidemic PPARα ligand, advanced the phase of activity onset by about 3 h independently of SCN in mice under LD condition. Therefore, PPARα might be associated with the advance of offset time of activity in the EE group observed in the present study. In contrast, because the HE hot-water extract did not show agonistic activity toward PPARα (4), it may be possible that factors other than PPARα play a role in the advance of the activity phase following WE intake.

In conclusion, EE only moved the offset time of the activity forward, whereas WE advanced the sleep–wake cycle without affecting the free-running period, photosensitivity, or clock gene mRNA expression levels in SCN. Our findings indicate that HE may serve as a functional food in the prevention and treatment of AD and DSPS.

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